Digital Implementation of Filters

EE 471C / EE 381K-17 Wireless Communication Lab

From Lecture #7

- The constellation energy and the transmit pulse shape energy are normalized
- The bandwidth of the transmitted pulse amplitude modulated signal is determined entirely from the pulse shape
- AWGN is the most common communication impairment
- Optimum pulse shapes satisfy certain special properties

Preview of Today's Lecture

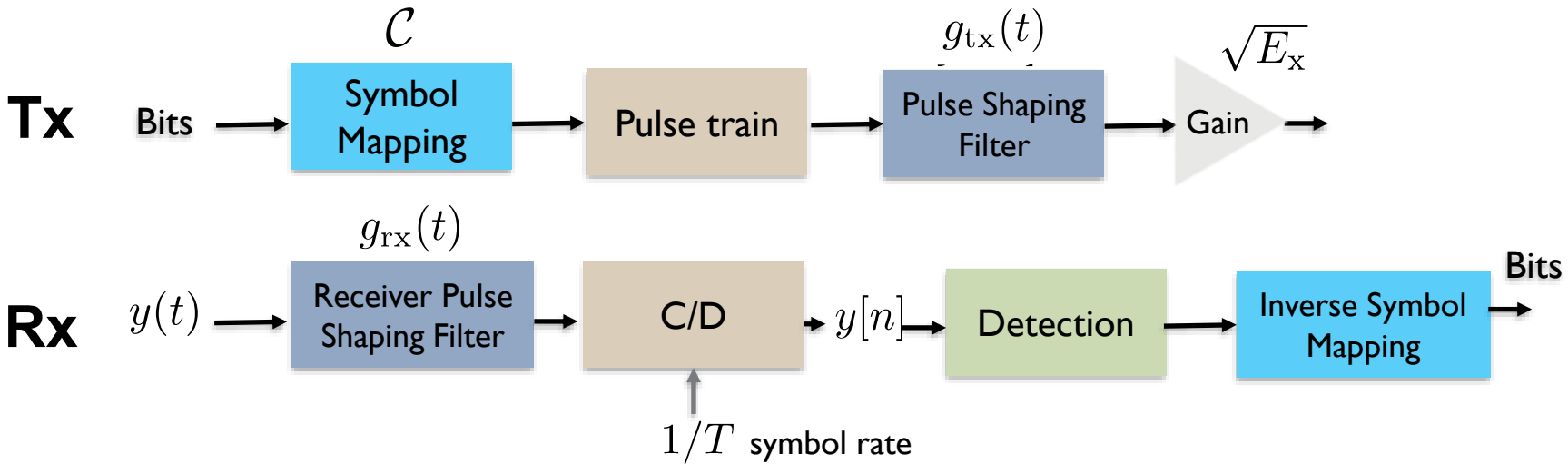
- Nyquist pulse shapes
  - Define Nyquist pulse criterion
  - Determine if a given pulse shape satisfies that criterion
- Digital implementation of transmit pulse shaping
  - Implement discrete-time pulse shaping at the transmitter using upsampling
  - Apply upsampling identities to implement and simplify transmit pulse shaping
- Digital implementation of receiver matched filter
  - Implement discrete-time pulse shaping at the receiver using oversampling combined with downsampling
  - Apply downsampling identities to implement and simplify receive pulse shaping

Nyquist Pulse Shapes

Learning objectives:

- Define Nyquist pulse criterion and determine if a given pulse shape satisfies that criterion.
- Describe key components of the transmitter and receiver

AWGN transmitter and receiver



- Problem: how to design $g_{\text{tx}}(t)$ and $g_{\text{rx}}(t)$ to maximize *SINR*

$$\text{SINR} = \frac{\text{S}}{\text{I}+\text{N}}$$

signal power

intersymbol interference

noise power

Pulse shape design

- The $g_{\text{rx}}(t)$that maximizes SNR is optimal
- Consider the received signal

$$y(t) = \sqrt{E_{\text{x}}} g_{\text{rx}}(t) \star g_{\text{tx}}(t) \star \sum_m s[m]\delta(t - mT) \; + \; g_{\text{rx}}(t) \star v(t)$$

- Let $g(t) = g_{\text{rx}}(t) \star g_{\text{tx}}(t)$
- After sampling

$$y[n] = \sqrt{E_{\text{x}}} s[n] + \sqrt{E_{\text{x}}} \sum_{m \neq n} s[m] g((n-m)T) \; + \; \tilde{v}[n]$$

noise

desired component

intersymbol interference

Signal-to-interference-plus-noise ratio

A good receiver maximizes the SINR

- The quality of the receiver can be measured by the signal-to-interference-plus-noise ratio (SINR)

$$SINR = \frac{E_{\mathrm{x}}|g(0)|^2}{N_{\mathrm{o}} \int |G_{\mathrm{rx}}(f)|^2 \mathrm{d}f \; + \; E_{\mathrm{x}} \sum_{m \neq n} |g(mT)|^2}$$

- Maximizing the SINR (through achievable upper bounds)

  (1) The received pulse shaping filter should be matched to the transmit pulse shaping filter

$$g_{\mathrm{rx}}(t) = g_{\mathrm{tx}}^{*}(-t)$$

  (2) The sampled composite pulse shaping filter should be a delta function

$$g(nT) = \delta[n]$$

Nyquist pulse-shapes (1)

- Zero ISI condition

$$g(nT) = \delta[n] \quad \Longleftrightarrow \quad \sum_k G\left(f + \frac{k}{T}\right) = T$$

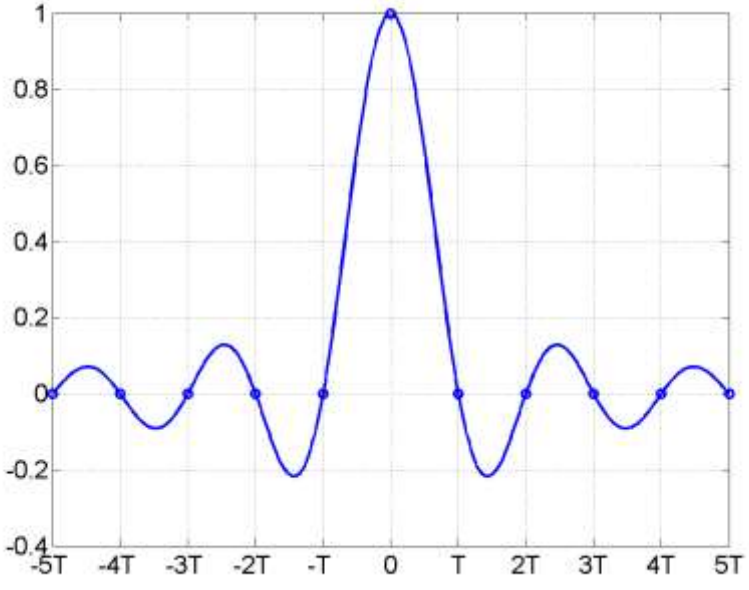
or equivalently with $g_d[n] \leftrightarrow G_d(e^{j2\pi f})$

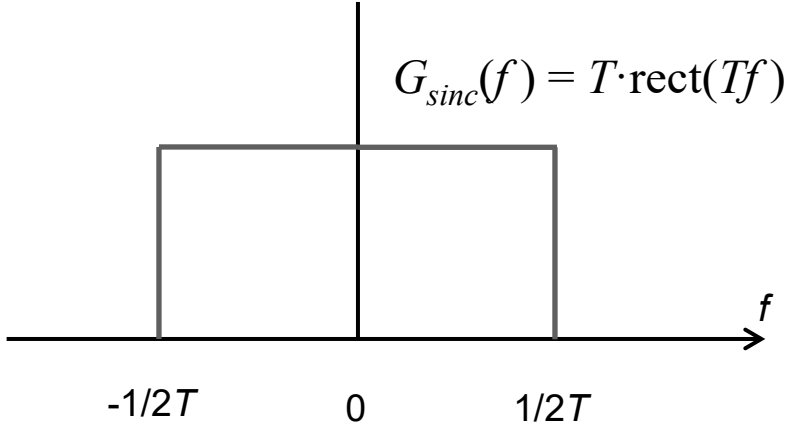
$$G_d(e^{j2\pi f}) = \sum_k G(fT + k) = 1$$

Nyquist pulse shapes (2)

- Example: sinc function

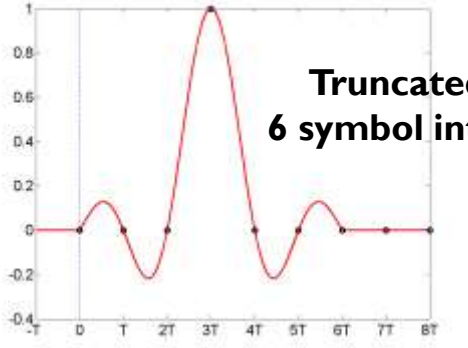
$$g_{\text{sync}}(t) = \frac{\sin(\pi t/T)}{\pi t/T}$$

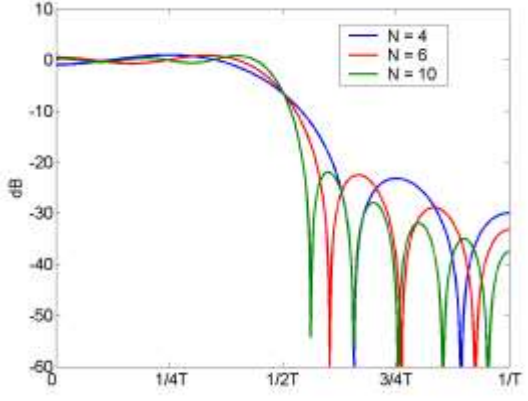




Nyquist pulse shapes (3)

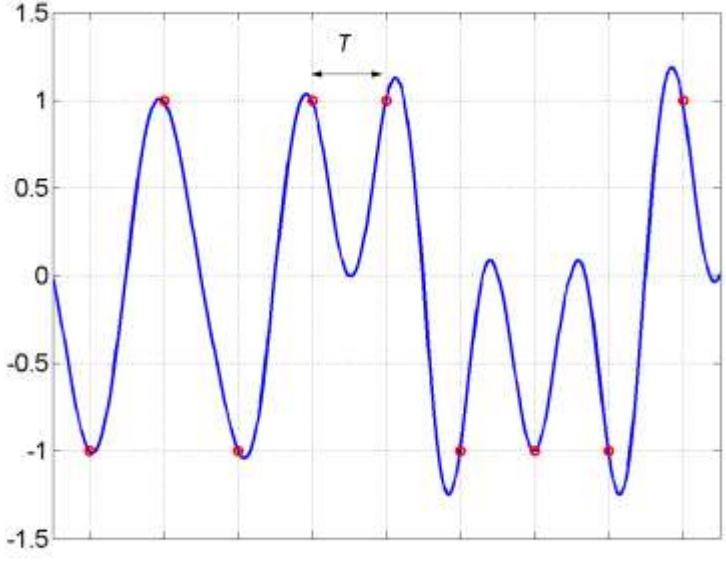
- The sinc pulse shaping filter has a number of problems in practice
  - Must be truncated
  - Is non-causal, but can be made causal through sufficient delay
  - Difficult to approximate steep band edges
  - Narrow main pulse implies lots of sensitivity to timing errors
  - Slow decay ~1/t





Nyquist pulse shapes (4)

- BPSK signal using a sinc pulse shaping filter
  - Pulse has been truncated to 20 symbol periods



  - Zero ISI

Example

Example 8 *Consider the function $g(t)$ with Fourier transform*

$$G(f) = \begin{cases} T, & \text{if } 0 \le |f| \le \frac{1-\alpha}{2T}, \\ T\left[\frac{1}{2} + \frac{1-2T|f|}{2\alpha}\right], & \text{if } \frac{1-\alpha}{2T} \le |f| \le \frac{1+\alpha}{2T}, \\ 0, & \text{if } \frac{1+\alpha}{2T} \le |f| \end{cases} \tag{59}$$

where α is the roll-off factor, $0 \le \alpha \le 1$.

1. *Is $g(t)$ a Nyquist pulseshape for symbol rate $1/T$, i.e., we assume that the signal $x(t)$ is sampled at the rate $1/T$?*

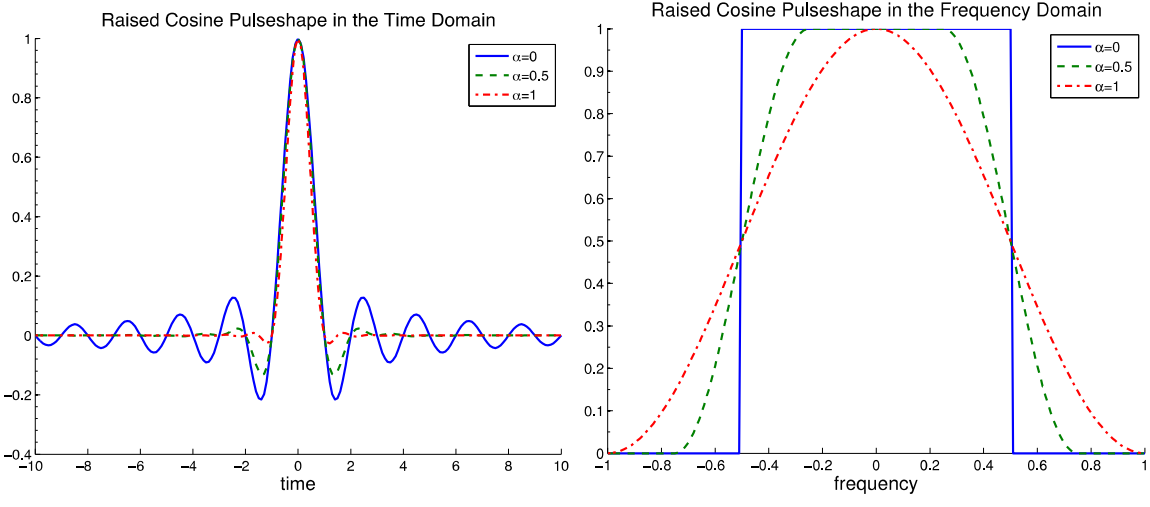
2. *Is $g(t)$ a Nyquist pulseshape for symbol rate $1/(2T)$?*

3. *Is $g(t)$ a Nyquist pulseshape for symbol rate $2/T$?*

Raised cosine pulse-shape (1)

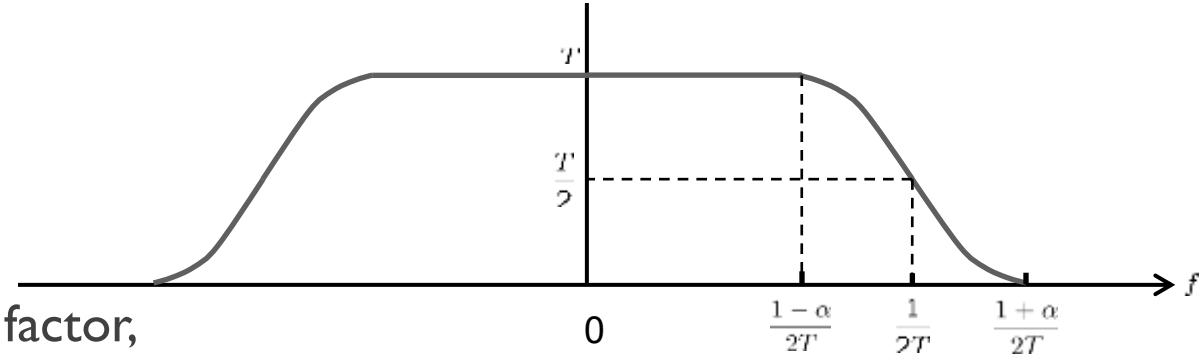
$$G_{\mathrm{rc}}(f) = \begin{cases} T, & 0 \le |f| \le (1-\alpha)/2T \\ \frac{T}{2}\left[1+\cos\frac{\pi T}{\alpha}\left(|f|-\frac{1-\alpha}{2T}\right)\right], & \frac{1-\alpha}{2T} \le \frac{1+\alpha}{2T} \\ 0, & |f| > \frac{1+\alpha}{2T} \end{cases}$$

$$g_{\mathrm{rc}} = \frac{\sin \pi t/T}{\pi t/T} \frac{\cos(\pi\alpha t)}{1-4\alpha^2 t^2/T^2}$$



Raised cosine pulse-shape (2)

- Features



  - α is the rolloff factor,
    - Often expressed as percentage excess bandwidth.
  - $(1+\alpha)/2T$ is the absolute bandwidth of the raised cosine pulse
  - $(1+\alpha)/T$ is the Nyquist rate of $x(t)$ with a raised cosine pulse shape
  - $1/T$ is the rate at which $x(t)$ is sampled
  - $\alpha = 0$ is the sinc pulse
  - Width of main lobe controlled by α
  - Decay at best ~ $1/t^3$

Square-root raised cosine

- Recall $G(f) = G_{\text{tx}}(f)G_{\text{rx}}(f)$ and for matched filter $G_{\text{rx}}(f) = G_{\text{tx}}^*(f)$
- Use the square-root raised cosine in practice for $g_{\text{tx}}(t)$ and $g_{\text{rx}}(t)$

$$g_{\text{sqrc}}(t) = \frac{4\alpha}{\pi\sqrt{T}} \frac{\cos\left((1+\alpha)\pi t/T\right) + \frac{\sin((1-\alpha)\pi t/T)}{4\alpha t/T}}{1-(4\alpha t/T)^2}$$

Noise with square-root raised cosine

$$C_{vv}[k] = \mathbb{E}_v\left[\tilde{v}[n]\tilde{v}^*[n+k]\right] \tag{77}$$

$$= \mathbb{E}_v\left[\int_{\tau_1}\int_{\tau_2} v(\tau_1)v^*(\tau_2)g_{\text{sqrc}}(nT-\tau_1)g_{\text{sqrc}}((n-k)T-\tau_2)\mathrm{d}\tau_1\mathrm{d}\tau_2\right] \tag{78}$$

$$= \sigma_v^2\int_{\tau} g_{\text{sqrc}}(nT_s-\tau)g_{\text{sqrc}}((n-k)T-\tau)\mathrm{d}\tau \tag{79}$$

$$= \sigma_v^2\int_{\tau} g_{\text{sqrc}}(\tau)g_{\text{sqrc}}(-kT-\tau)\mathrm{d}\tau \tag{80}$$

$$g_{\text{sqrc}}(t) = g_{\text{sqrc}}(-t)$$

$$C_{vv}[k] = \sigma_v^2\int_{\tau} g_{\text{sqrc}}(\tau)g_{\text{sqrc}}(kT+\tau)\mathrm{d}\tau.$$

$$g_{\text{rc}}(t) = \int_{\tau} g_{\text{sqrc}}(t-\tau)g_{\text{sqrc}}(\tau)\mathrm{d}\tau$$

$$= \int_{\tau} g_{\text{sqrc}}(t+\tau)g_{\text{sqrc}}(\tau)\mathrm{d}\tau$$



$$C_{vv}[k] = \sigma_v^2 g_{\text{rc}}(kT)$$

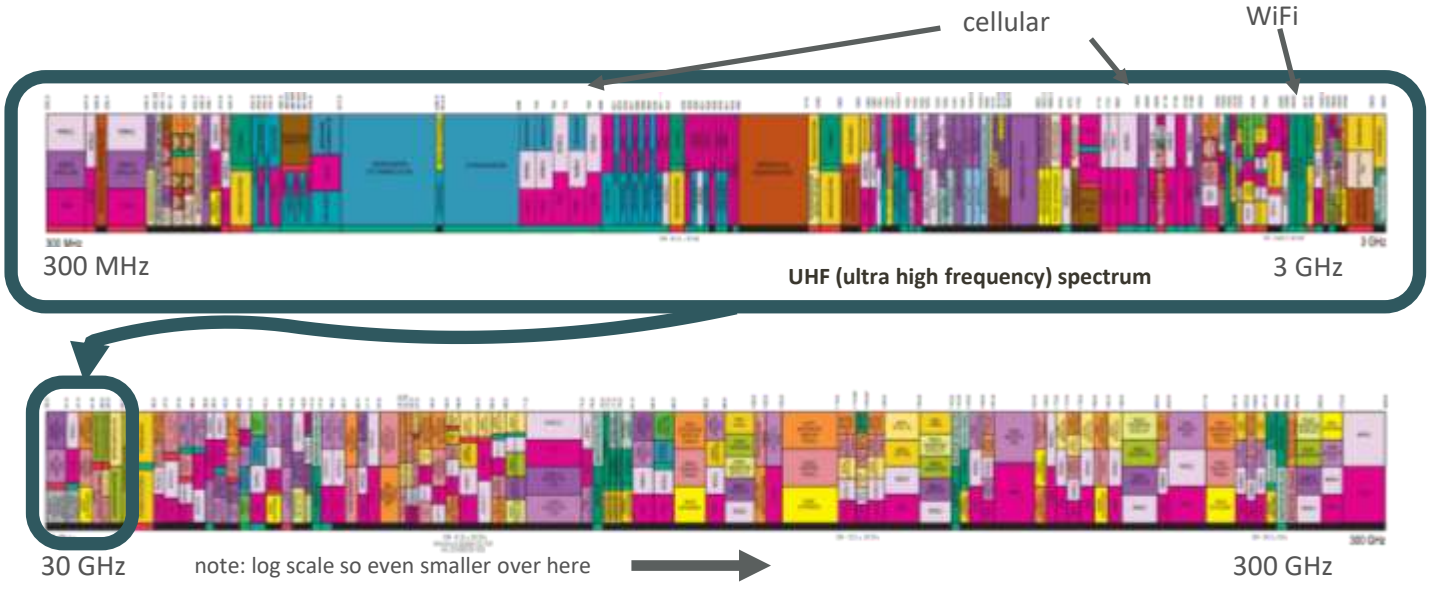
$$= \sigma_v^2\delta[k]$$

Nyquist Pulse Shapes

Learning objectives:

- Define Nyquist pulse criterion and determine if a given pulse shape satisfies that criterion.
- Describe key components of the transmitter and receiver

Millimeter Wave Spectrum



- Lots of spectrum in the mmWave bands (large bandwidths)
- Used by next generation IEEE 802.11ad, cellular (??)

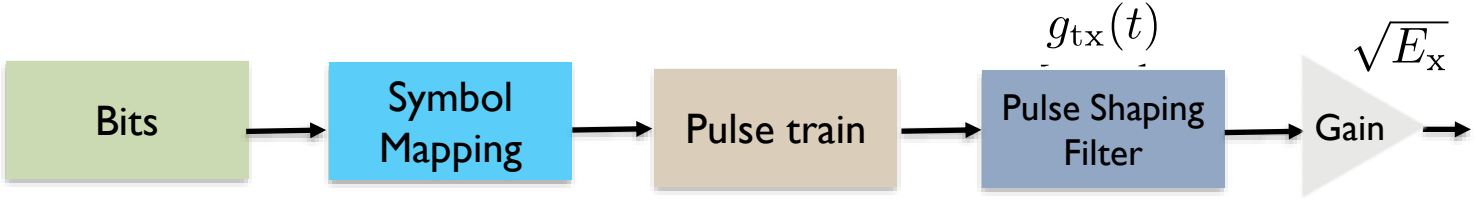
Digital Implementation of Transmit Pulse Shaping

Learning objectives:

- Implement discrete-time pulse shaping at the transmitter using upsampling.
- Apply upsampling identities to implement and simplify transmit pulse shaping

System model for the transmitter

- Consider the following model for the transmitter



- This structure does not map to a realizable implementation
  - Mix of analog and digital implementations
  - Would like to do as much as possible in digital
  - There is no discrete-to-continuous (D/A) converter
  - $g_{tx}(t)$ may have bandwidth greater than $1/(2T)$

Generation of the transmitted signal (1)

- We focus on generating the transmit signal prior to scaling

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} s[n] g_{\text{tx}}(t - nT)$$

- $x(t)$ is bandlimited, so according to the Nyquist sampling theorem

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} c[n] \text{sinc}((t - nT_x)/T_x)$$

  - T_x is the sampling period
  - $1/T_x$ is greater than twice the maximum frequency of $g_{\text{tx}}(t)$
  - $c[n] = \tilde{x}(nT_x)$

$$= \sum_{m=-\infty}^{\infty} s[m] g_{\text{tx}}(nT_x - mT)$$

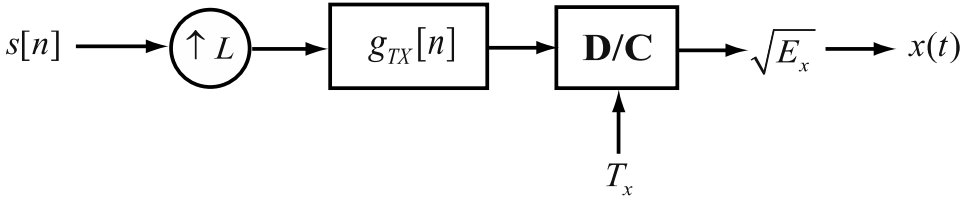
How to create this discrete time signal?

Generation of the transmitted signal (2)

- Generation of $c[n]$
  - Suppose that $T_x = T/L$ for some integer L and let $g_{\text{tx}}[n] = g_{\text{tx}}(nT_x)$
  - Then

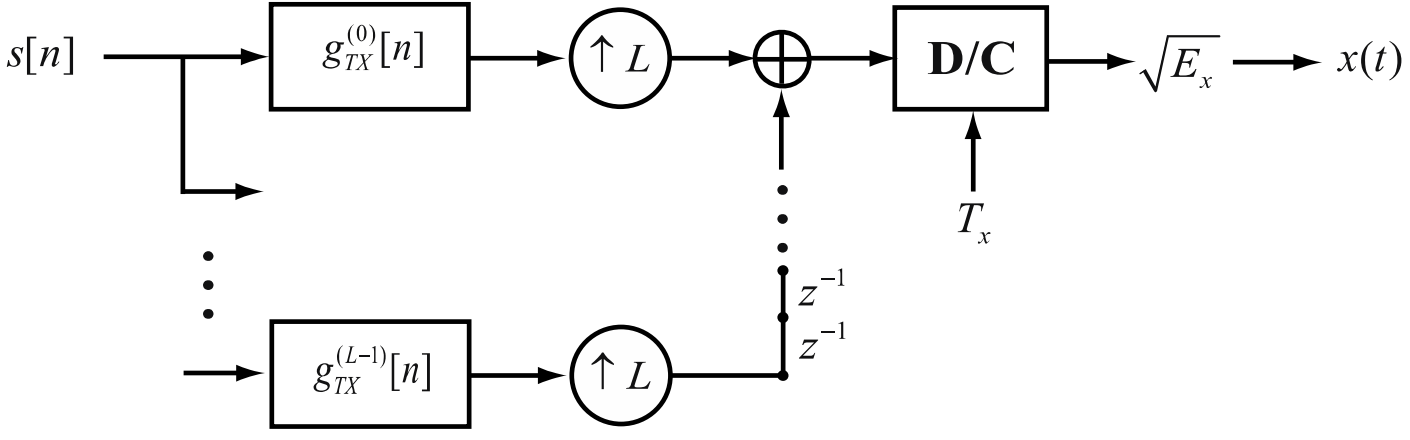
$$c[n] = \sum_{m=-\infty}^{\infty} s[m] g_{\text{tx}}[n - Lm] = \left(\sum_{m=-\infty}^{\infty} s[m] \delta[n - Lm] \right) \star g_{\text{tx}}[n]$$

  - This can be implemented using upsampling $\uparrow L$ operation
    - Upsampling by L corresponds to inserting $L - 1$ zeros after every $s[n]$



Efficient implementation of transmit pulse shaping

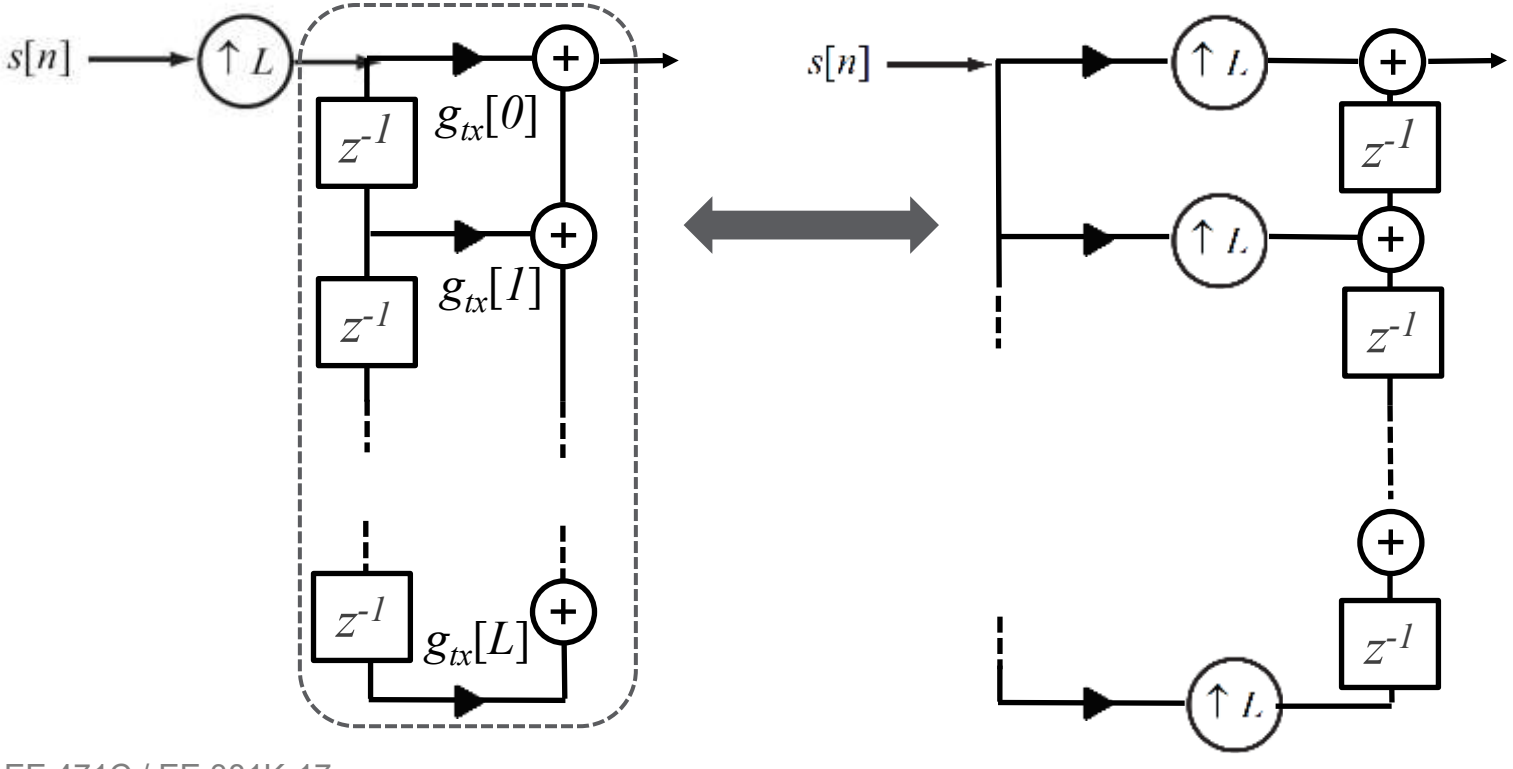
- Convolution after upsampling does not account for the many zeros at the input
- A more efficient approach is to use a filter bank
  - Filtering is done over the non zero samples



- How to construct $g_{tx}^{(k)}[n]$, $k=0,\ldots,L-1$ from $g_{tx}[n]$?

Polyphase decomposition

- The underlying idea is to apply the following identity



Polyphase decomposition

- The ℓth polyphase component of $c[n]$ is

$$\begin{aligned} c^{(\ell)}[n] &= c[nL+\ell] \\ &= \sum_{m=-\infty}^{\infty} s[m] g_{\text{tx}}[nL+\ell-mL] \end{aligned}$$

- The polyphase components of the transmit pulse shaping filter are

$$g_{\text{tx}}^{(\ell)}[k] = g_{\text{tx}}[kL+\ell]$$

- $c[n]$ can be reconstructed as

$$c[n] = \sum_{\ell=0}^{L-1} \sum_{k} \left(\sum_{m=-\infty}^{\infty} s[m] g_{\text{tx}}^{(\ell)}[k-m] \right) \delta[n-kL-\ell]$$

Digital Implementation of Transmit Pulse Shaping

Learning objectives:

- Implement discrete-time pulse shaping at the transmitter using upsampling.
- Apply upsampling identities to implement and simplify transmit pulse shaping

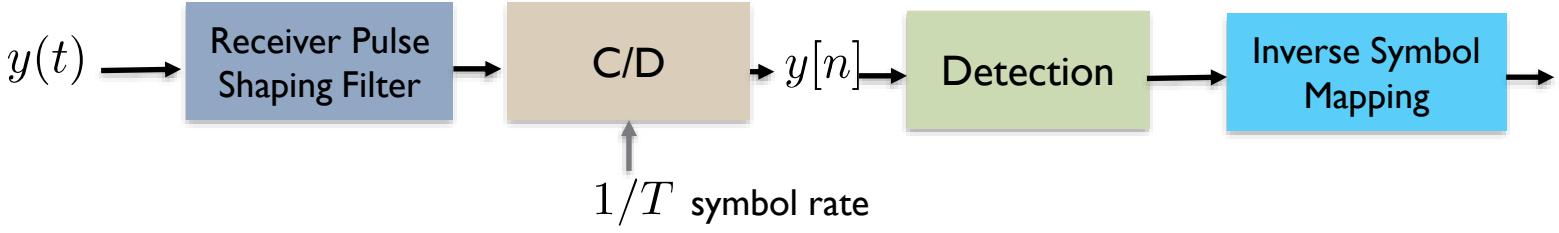
Digital Implementation of Receiving Matched Filter

Learning objectives:

- Implement discrete-time pulse shaping at the receiver using oversampling combined with downsampling
- Apply downsampling identities to implement and simplify receive pulse shaping

System model for the receiver

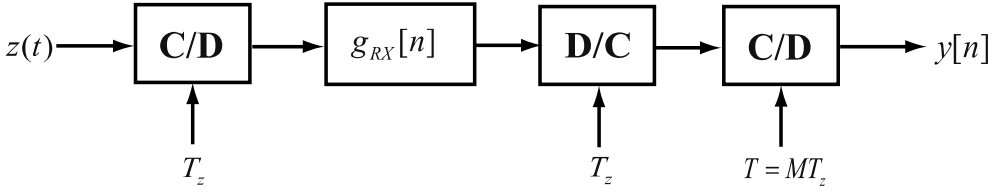
- Consider the following model for the receiver



- This structure does not map to a flexible implementation
  - Requires analog matched filtering
  - Does not take advantage of flexible digital processing
  - Does not lend itself to various synchronization tasks
- How can we implement pulse-shaping in discrete time?

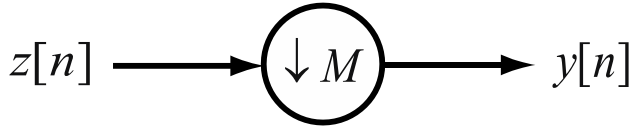
Digital implementation of matched filter

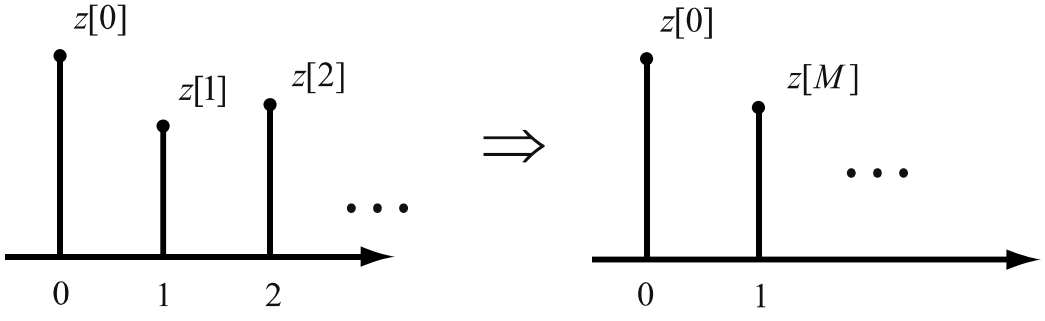
- Use discrete-time processing of a band limited continuous-time signal
  - Implement the analog filter using discrete-time processing



  - Let $T_z = T/M$ for some integer M. If $1/T_z >$ Nyquist rate, then this is called oversampling
  - $g_{\mathrm{rx}}[n] = g_{\mathrm{rx}}(nT_z)$
  - Simplify using downsampling by M

Downsampling





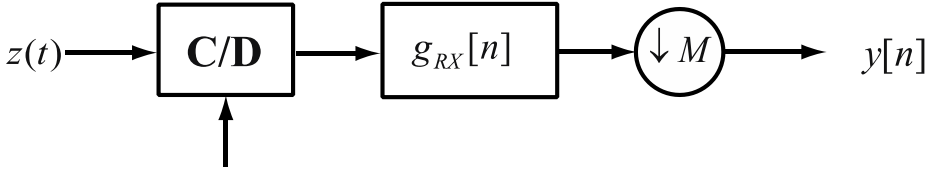
$$y[n] = z[nM]$$

$$Y(e^{j2\pi f}) = \frac{1}{M}\sum_{m=0}^{M-1} Z\left(e^{j2\pi f/M - j2\pi m/M}\right)$$

Digital implementation of matched filter

- If $z[n] = z(nT_z)$ we can simplify with downsampling

$$y[n] = \sum_k g_{\text{rx}}[k] z[nM - k]$$



- Lots of filtering computations are still wasted because we discard those coefficients when downsampling

Digital implementation of matched filter

- Use downsampling equivalence to reduce computations

$$
\begin{aligned}
y[n] &= \sum_m z[mM]g[n-m] \\
&= \sum_m z[m]\left(\sum_p g[p]\delta[nM-m-pM]\right) \\
&= \sum_m z[m]\bar{g}[nM-m]
\end{aligned}
$$

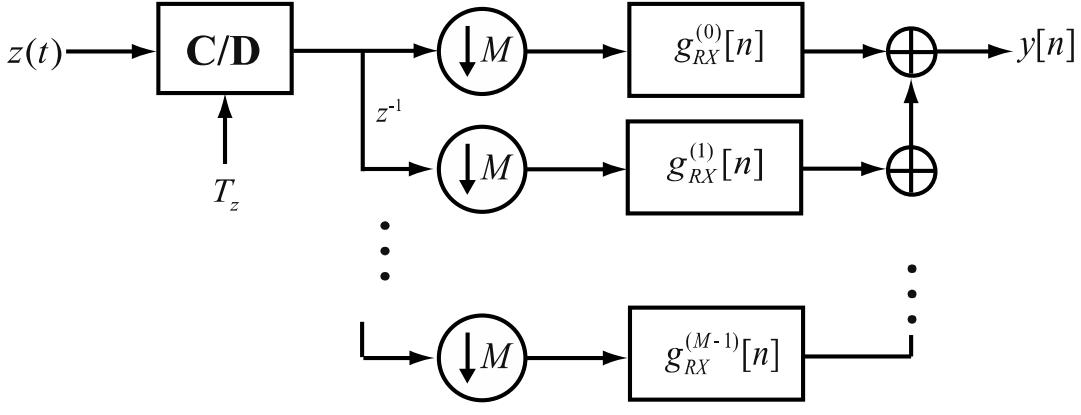
Digital implementation of matched filter

- Use a transformation $k = pM - m$

$$
\begin{aligned}
y[n] &= \sum_k g_{\text{rx}}[k] z[nM-k] \\
&= \sum_k z[k] g_{\text{rx}}[nM-k] \\
&= \sum_p \sum_{m=0}^{M-1} z[pM-m] g_{\text{rx}}[nM-pM+m] \\
&= \sum_k \sum_{m=0}^{M-1} z[pM-m] g_{\text{rx}}^{(m)}[n-p]
\end{aligned}
$$

Digital implementation of matched filter

- We can finally exchange of downsampling and filtering operations



Digital Implementation of Receiving Matched Filter

Learning objectives:

- Implement discrete-time pulse shaping at the receiver using oversampling combined with downsampling
- Apply downsampling identities to implement and simplify receive pulse shaping